

■ DrugLens

AI-Powered Geriatric Polypharmacy Safety Auditor

Track: Unicorn Track (AMD Developer Hackathon: Act II)

Built with: local MedGemma 4B + TxGemma 2B on AMD Instinct GPUs, Gemma 4 in cloud

The Geriatric Polypharmacy Crisis

Aging pharmacokinetics and polypharmacy create high-stakes clinical risks:

- **Polypharmacy Epidemic:** Over 40% of seniors take 5+ daily prescription drugs.
- **Severe Adverse Drug Events (ADEs):** Causes 3.5 million doctor visits and 125,000 hospitalizations annually.
- **Renal/Hepatic Decline:** Decreased clearance rates lead to drug accumulation and toxicities.
- **High System Costs:** Avoidable ADEs incur over \$3.5 Billion in yearly emergency care costs.

Why Current EHR Systems Fail

Existing electronic records and clinical checkers have critical shortcomings:

- **Severe Alert Fatigue:** Checkers flag every interaction, causing doctors to ignore 90% of warnings.
- **Demographic Blindspot:** Standard checkers ignore patient age, eGFR, or disease contraindications.
- **Unstructured Clinical Text:** Medical notes are written in free-text clinical prose, not clean tables.
- **Novel Drug Blindspot:** Lookup databases fail on undocumented or newly launched drug pairs.

The Solution: DrugLens

DrugLens combines clinical rulesets with a multi-model Gemma pipeline running on AMD GPUs:

- **Natural Language Parsing:** Local MedGemma extracts structured prescriptions from raw clinical text.
- **Verified Clinical Auditing:** Immediate local check against 2023 AGS Beers Criteria & STOPP/START v3.
- **Molecular DDI Prediction:** Local TxGemma evaluates SMILES strings to predict novel interactions.
- **Professional Synthesis:** Configurable cloud LLM writes doctor narratives and patient summaries.

Multi-Model Pipeline on AMD Instinct

Optimal partitioning of clinical workloads secures patient privacy and maximizes compute speed:

MedGemma 4B Local AMD GPU (vLLM)	TxGemma 2B Local AMD GPU (vLLM)	Gemma 4 31B Fireworks AI Cloud
Clinical Text Parser Extracts drug names, route, dose, and frequencies from raw clinical notes.	Molecular DDI Predictor Predicts novel structural interactions using PubChem molecular SMILES strings.	Report Orchestrator Drafts professional clinical summaries and patient explanations in plain English.

Resilient Fallback Design

The system degrades gracefully when local GPU resources or cloud endpoints are offline:

Pipeline Stage	Primary Engine (AMD Instinct GPU)	Graceful CPU/Local Fallback
Medication Parsing	MedGemma 4B-it (Local vLLM)	Structured Regex parsing engine
Drug Interactions (DDI)	TxGemma 2B-it (Local vLLM)	Verified local database matching (100+ pairs)
Narrative Synthesis	Gemma 4 31B (Fireworks API)	Rule-based scoring & template rendering

Summary: Why DrugLens Wins

- **AMD Synergy:** Uses local AMD Instinct GPUs for private, high-performance clinical workloads.
- **Rigorous Clinical Foundations:** Match against AGS Beers and STOPP/START rules minimizes hallucinations.
- **No Hallucinations on Known Rules:** High-risk clinical rules are matched deterministically.
- **No Alert Fatigue:** Intelligent model routing focuses clinician attention on critical cases first.
- **Ready for Integration:** Fast, lightweight containerized FastAPI/React stack is fully cloud-deployable.